

OncoVI without OncoKB: 7802 MTB variants

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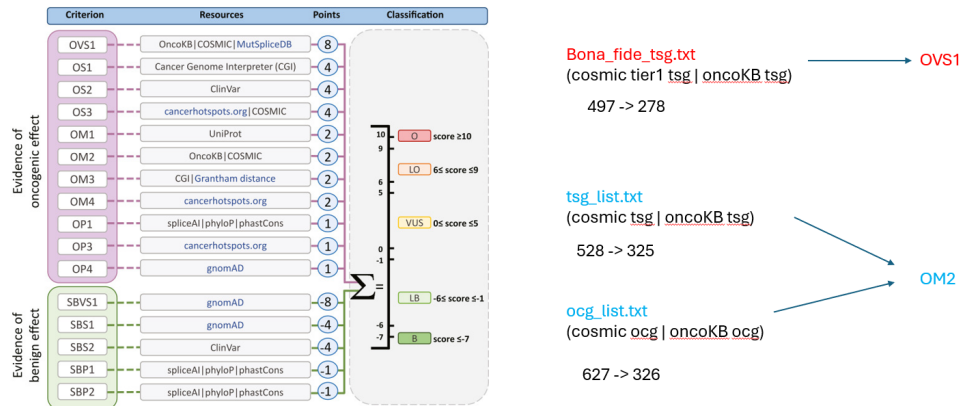
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1 Introduction

OncoVI (*Carta et al., Journal of Molecular Diagnostics, 2026*) integrates information from external databases subjected to licensing term such as COSMIC and OncoKB. OncoVI uses these resources to define the role of genes as tumor suppressor genes (TSGs) or oncogenes (OGs) for the evaluation of specific criteria. Here, we want to define the role of TSGs and OGs relying on COSMIC only and assess how OncoVI score and classification change between OncoVI (with OncoKB) and OncoVI_mod (not relying on OncoKB). Data used for the comparison between the two versions: Molecular Tumour Board (MTB) set of 7,802 variants detected in more than 500 tumors analysed with the Illumina targeted gene panel TSO500 (analysed in OncoVI paper).



Adapted from Figure 1, *Carta et al., Journal of Molecular Diagnostics, 2026*.

Number of genes exclusive to OncoKB database and therefore missing in OncoVI_mod compared to OncoVI:

“bona_fide_tsg.txt”: 219

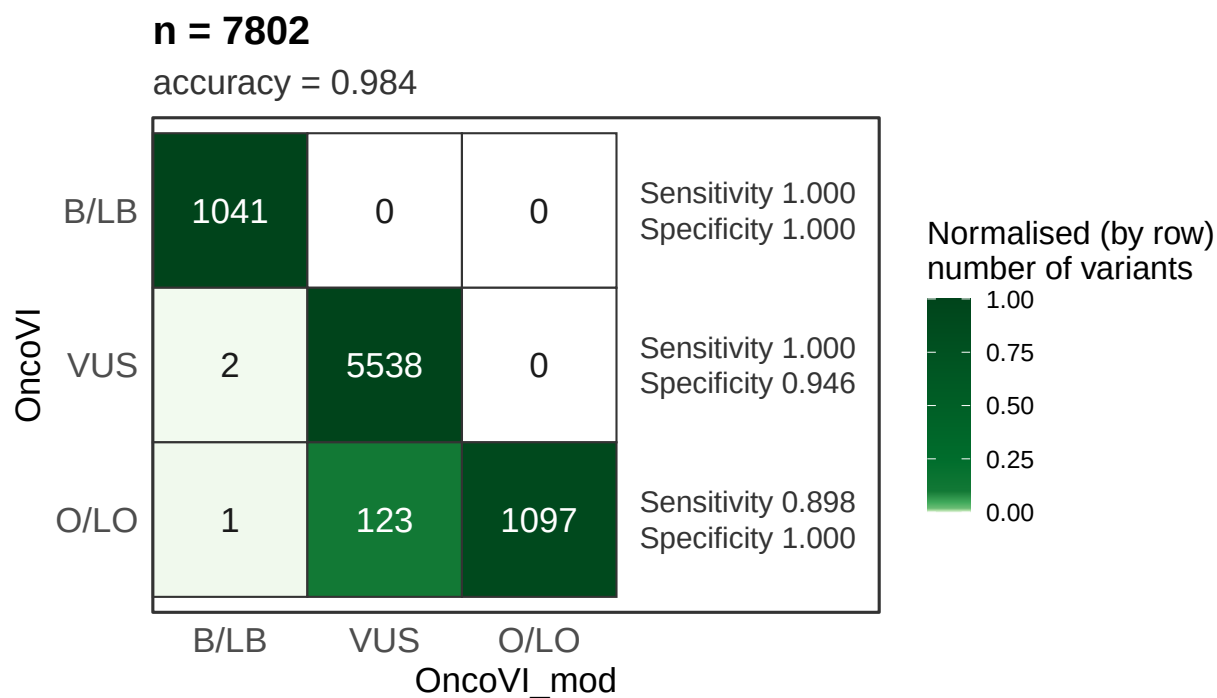
“tsg_list.txt”: 203

“ogs_list.txt”: 301

2 Results

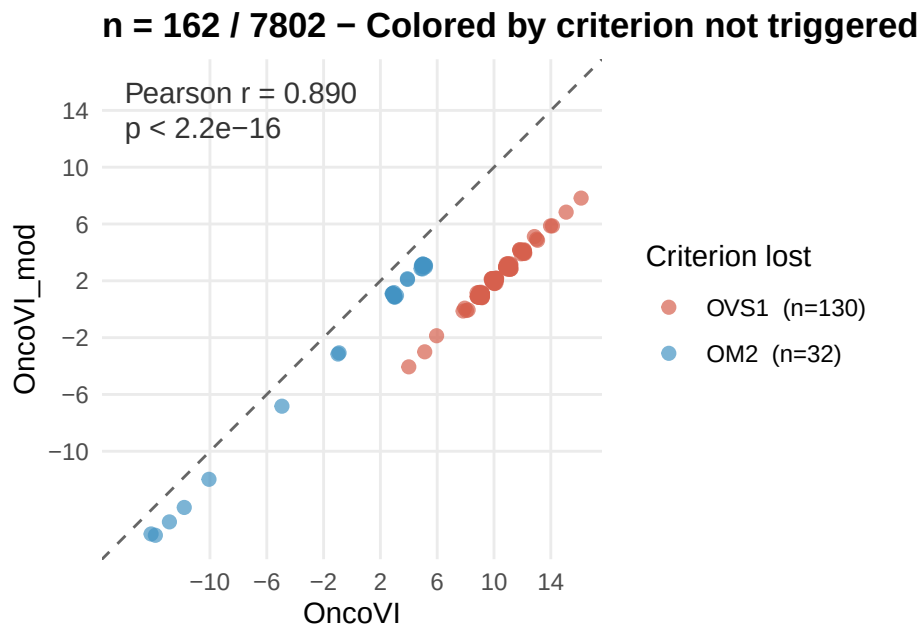
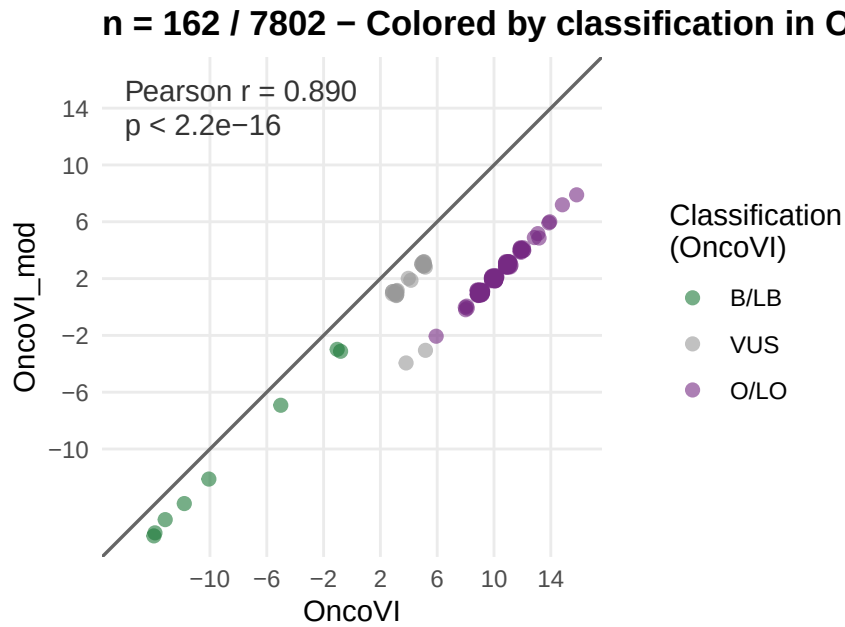
2.1 7802 MTB variants

2.1.1 Confusion matrix



Out of the total of 7802 variants, for 126 variants the classification changed, and for 162 variants the scores changed.

2.1.2 Variants with OncoVI score different between OncoVI and OncoVI_mod

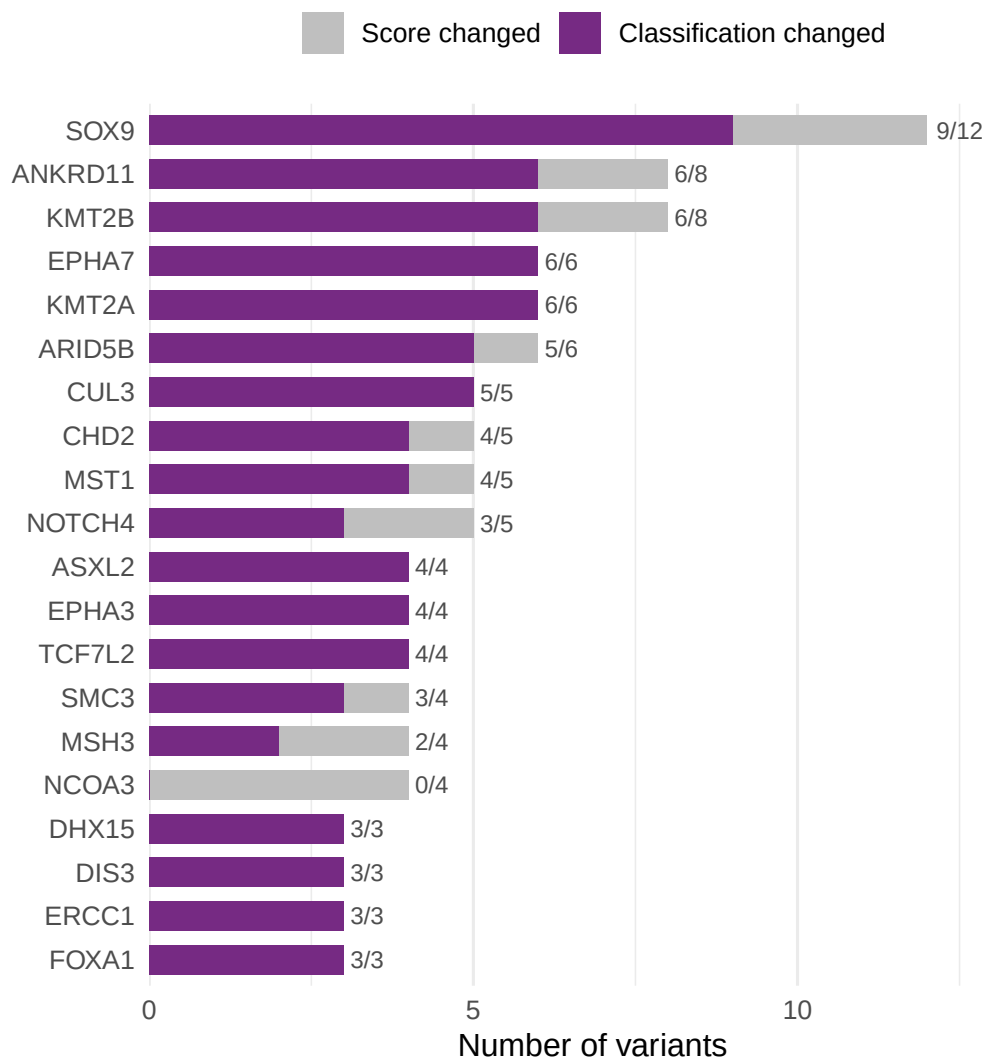


When the score of a variant changes in OncoVI_mod, it is always lower than in OncoVI, as either OVS1 or OM2 criterion is not triggered in OncoVI_mod.

2.1.3 Genes most frequently affected

The 162 variants with score changed between OncoVI and OncoVI_mod belong to 59 different genes. Top affected genes:

Top 20 genes



In summary, we evaluated OncoVI_mod using the 7,802 somatic variants from the Molecular Tumour Board (MTB) data set, comprising variants detected in more than 500 tumours analysed with the Illumina TruSight Oncology 500 (TSO500) gene panel. We compared the results obtained with OncoVI (which relies on OncoKB) and OncoVI_mod (not relying on OncoKB).

Overall, the two configurations showed high agreement, with an accuracy of 0.984. The oncogenicity classification changed for 126 of 7,802 variants, while the variant-specific score changed for 162 variants. For the variants with a changed score, the score was consistently lower in the configuration without OncoKB and was associated with the loss of either the OVS1 or OM2 criterion.